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Capstone Project

ITAI-3377 A.I at the Edge

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Generative Fashion Micro Factory

1. Objective

Edge AI promises to collapse the distance between imagination and manufacturing. For this capstone I'm proving that promise inside a domain I love, fashion! Through designing a suitcase sized "micro factory" that stitches custom made garments right where people need them. The concept is entirely hypothetical. Every diagram, test, and metric is forward looking; yet I'm hoping the architecture is detailed enough that an engineering team could start building tomorrow. If it works, a rain soaked aid worker, a festival goer short on layers, or a soldier in need of mission specific gear could walk up, describe a jacket, and pull it on two hours later. What follows is the complete blueprint: goals, system wiring, AI models, zero trust security, and a simulation only testing plan.

2. Project Proposal

2.1. Use Case

Imagine a pop up booth at a music festival, a forward military camp that just soaked its uniforms in the rain, or a disaster relief hub where children need warm clothes now. The Generative Fashion Micro Factory (GFMF) gives those locations a mini textile factory on wheels. A customer walks up and says, "I need a lightweight rain jacket in olive drab, medium long torso." I (or a single attendant) guide them through a quick body scan, let the system do its thing, and hand them the jacket before their next shift or concert set.

2.2. Generative AI ingredients

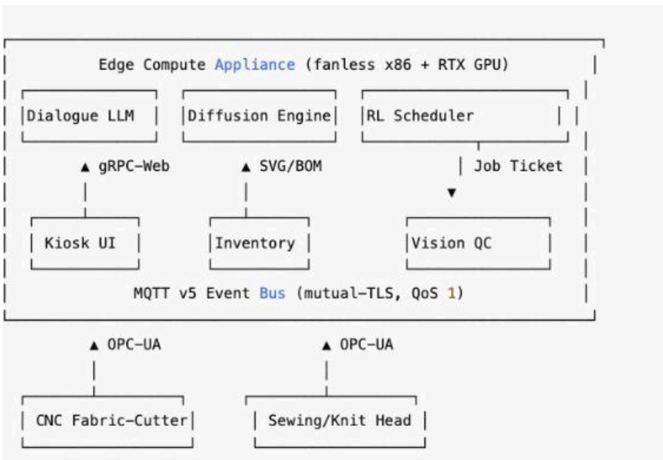
- *Diffusion Pattern Engine* – A Stable Diffusion style network, fine tuned on ~100 apparel CAD files, converts my text/voice prompt + optional sketch into vector seam patterns (SVG) and a bill of materials (BOM).
- *Conversational LLM Agent* – A 3 billion parameter transformer, distilled and 4-bit quantised, conducts the design interview in a friendly, "fashion consultant" tone until the prompt is precise enough for the diffusion engine.
- *RL Scheduler* – A reinforcement learning agent slots the job onto available machines, chooses the optimal fabric roll and nests pattern pieces to shrink the excess waste.

2.3. Targets and Outcomes

- *Speed*: Concept-to-cloth ≤ 2 h; ≥ 12 garments per 10 hour shift.

- *Sustainability*: ≥ 15 % less off cut waste; zero deadstock. Local production kills freight emissions.
- *Experience*: One operator, zero queue chaos, fully crafted garments on demand.

3. Inside the System



The entire stack is housed in a rugged carry on case. When opened the two industrial devices automatically connect over OPC-UA, and the kiosk tablet launches a chat window.

3.2. Walk through of a Garment

1. *"Hello, Inspiration"* :I chat with the booth patron; their voice/text flows over gRPC-Web (HTTP/2 + TLS) to the LLM.
2. *Prompt Lock-in*: When the LLM is confident it has style, size, climate, and aesthetic nailed, it emits a JSON block on MQTT topic pattern/request.
3. *Magic Pattern Generation*: The Diffusion Engine picks up the request, grabs the customer's encrypted body scan profile (optional), and spits out jacket.svg + bom.yaml in ~1's.
4. *Smart Scheduling*: The RL Scheduler subscribes to pattern/complete, checks BLE inventory tags, nests pieces on the shortest fabric roll, and posts a signed job ticket (job/new).
5. *Cut & Sew:* Cutter and sewing station authenticate with TPM bound certs, pull G-code over OPC-UA, and start humming.
6. *Quality Check*: A 60 fps camera streams RTSP; a lightweight auto encoder flags seam anomalies. Pass? Publish job/complete. Fail? Loop back for rework.
7. *Done & Dusted*: The kiosk prints a QR care label; encrypted usage stats queue for a batch HTTPS upload every 30 min.

4. Inside the AI

4.1. Diffusion Pattern Engine

- *Model:* Latent diffusion U-Net, cross attention on style tokens (garment type, climate, fabric weight).
- *Data:* 80 % open CAD patterns, 20 % proprietary scans; every sample tagged with body size metadata so plus size patrons aren't an afterthought.
- *Edge tuning:* INT8 TensorRT conversion keeps single sample inference under a second on an RTX A5000 Mobile.

4.2. Conversational LLM

- *Skeleton:* Distilled GPT-J-6B → 3 B params, quantised to 4-bit blocks; streamed with llama.cpp.
- *Personality:* System prompt sounds like a helpful stylist. "Runway mentor meets engineering nerd."
- *Independent guardrails*: The LLM stops once it outputs a schema valid JSON; no rogue instructions or infinite chatter.

4.3. RL Scheduler

- *State vector:* Fabric length per roll, estimated cut time, defect probability history, machine queue depth.
- *Actions*: Pick roll, nesting offset, machine order, time slot.
- *Reward*: 0.7 \times \text{waste}^{-1} + 0.3 \times \text{throughput}.
- *Training*: Proximal Policy Optimisation in simulation; heuristic fallback if confidence dips below 0.8.

4.4. (Imaginary) Toolchain

All models live in PyTorch during R&D, export to ONNX, then land on the edge via TensorRT. For low resource hardware I'd swap in OpenVINO or TensorFlow Lite but that's a future sprint.

5. Security & Ethics: My Non-Negotiables

5.1. Zero trust from plug-in-power

- TPM anchored device certificates; SPIFFE/SPIRE hands out short lived IDs.
- Mutual TLS on every pipe: MQTT, OPC-UA, gRPC.
- Model bundles are signed; SHA-256 chain verified before hot swap.

5.2. Privacy, Please

Body scans live encrypted on the NVMe, are deleted 24-h after pickup (unless the customer opts in for future tailoring), and never touch the public cloud. Aggregate metrics get a dash of Laplacian noise so even it can't be traced back to a single wearer.

5.3. Bias & cultural respect

- Audit the CAD dataset for under representation of non standard sizes and up weight those rows.
- If someone tries to replicate a protected cultural garment, the LLM pops a friendly note: "Heads-up, this design is culturally significant. Want to learn more?"

6. How I'd roll it out (if I were coding)

6.1. Day 1 deployment sequence

- *Secure boot handshake:* I tap an NFC hardware root card; the appliance validates the root of trust.
- *Device discovery:* Cutter and sewer broadcast OPC-UA FindServers; the appliance whitelists them once certs match.
- *Model flash:* Encrypted USB key copies the diffusion and LLM weights; signatures check out.
- *Smoke test:* The scheduler runs a scrap fabric mini pattern, camera verifies stitches, and a green LED screams "Ready!"

Test block	Tool / sim	Pass metric
Functional twin	SimPy event model	≥ 95 % of simulated jobs ≤ 2 h
Load/stress	Locust (50 kiosk sessions, 10 k msg s^{-1})	Pattern-req → job-ticket p95 ≤ 250 ms
Security red team	MITRE Caldera	No unauthorised MQTT publish; OPC-UA escalation blocked
Fail-over	Pull WAN for 1 h	Production loop continuous; backlog auto-syncs
Disaster recovery	Hard power-cycle mid-cut	Job resumes; fabric loss ≤ 5 %

Everything runs in Docker Compose, enabling replicas to spin up in minutes; Grafana dashboards track latency, waste, and defect trends without sacrificing any physical cloth during testing.

7. Challenges & Trade-Offs

- Edge hardware cost vs. privacy wins: GPUs aren't cheap, but they keep PII off the cloud and latency microscopic.
- Lint induced cooling drama: Sewing kicks up fibers; need positive pressure, filtered airflow or the fanless chassis will choke.
- MQTT's simplicity vs. AMQP's muscle MQTT is feather light; missing fancy routing but gaining footprint and brainspace.
- Dataset bias cleanup: Early trial runs mis sized plus size bodies; augmenting with synthetic scans shaved sizing error by a percentage..
- Big lesson: By threat modeling before sketching the network, embedded security instead of bolting it on later.

8. Wrapping Up & Next Moves

If someone green lights this concept, here's my personal roadmap:

- **Next 90 days:** Train a "mini diffusion" on ten garment classes, bolt it to a hobby grade CNC cutter and validate the control loop.

- **Six month sprint**: Upgrade to industrial hardware, refine the RL scheduler in live queue conditions, and pilot a public booth at a maker faire or field exercise.

Conclusion:

Ultimately, the aim is both clear and ambitious: transform a spark of inspiration into a finished garment exactly when and where it's needed without producing a gram of excess waste. By fusing generative AI, edge computing, and industrial IoT inside a carry on sized micro factory, this concept converts apparel creation from a slow, centralized supply chain into an agile, hyper local service. The payoff is faster relief in humanitarian settings, lighter logistics for deployed teams, and richer personalization for everyday consumers. All delivered with airtight data protections and true zero waste production.

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